

Problem B, Cell 2: D_B at triangular n

$$D_B(T_k) = k^2 - k \text{ for every } k \geq 1$$

Mathematics research team (task B-C2), written up by the Scribe

Cell status: SOLVED. **Claim status:** PROVED.

Answer. For every $k \geq 1$,

$$D_B(T_k) = k^2 - k \quad (\text{no exceptions; } k = 1 \text{ gives } 0, k = 2 \text{ gives } 2).$$

Explicit extremal partition. $\lambda^{(1)} = (1)$ and, for $k \geq 2$,

$$\lambda^{(k)} = (k-1, k-1, k-2, \dots, 2, 1, 1),$$

i.e. $\lambda_1 = k-1$, $\lambda_j = k+1-j$ for $2 \leq j \leq k$, $\lambda_{k+1} = 1$; it satisfies $d_B(\lambda^{(k)}) = k^2 - k$. Both bounds are proved in full in Section 4; no cited result is used as a proof step.

1 Statement

Definitions (verbatim from the official statement). A *partition* of $n \geq 1$ is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with sum n (s piles). The *shift* $B(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . λ is *cyclic* if $B^i(\lambda) = \lambda$ for some $i \geq 1$.

$$d_B(\lambda) = \min\{i \geq 0 : B^i(\lambda) \text{ is cyclic}\}, \quad D_B(n) = \max\{d_B(\lambda) : \lambda \text{ a partition of } n\}.$$

$$T_k = k(k+1)/2, \delta_k = (k, k-1, \dots, 2, 1).$$

Cell text (verbatim). “Next, how long the process can take to reach a cycle, starting with the triangular numbers. Determine $D_B(T_k)$ for every k , with proof of both bounds.”

Exact target.

- Give an explicit formula $F(k)$ (with any small- k exceptions stated separately and exactly) such that $D_B(T_k) = F(k)$ for *every* $k \geq 1$.
- UPPER bound: prove $d_B(\lambda) \leq F(k)$ for every partition λ of T_k , for every $k \geq 1$.
- LOWER bound: give an explicit partition $\lambda^{(k)}$ of T_k , as a function of k , and prove $d_B(\lambda^{(k)}) = F(k)$ (in particular $\geq F(k)$) for every $k \geq 1$.
- Any fact about which partitions of T_k are cyclic must itself be proved in the submission.

Hand-in: a written proof (LaTeX), stating clearly what is established. Computation may be used only if exhaustive over a finite set the argument has reduced the problem to (code included, < 10 min); small- k tables alone prove nothing for all k .

2 What is established

Claim	Status / where
$F(k) = k^2 - k$ for all $k \geq 1$, no exceptions: $D_B(T_k) = k^2 - k$	PROVED; Parts A, B
Shift on diagrams = diagonal rotation, plus left-slide of rows $> s$ when $s < \lambda_1 - 1$	PROVED; 0.1
δ_k is fixed by B	PROVED; 0.2
Every orbit of T_k reaches δ_k ; δ_k is the only cyclic partition of T_k	PROVED; 0.3
$\lambda^{(k)}$ has $d_B = k^2 - k$ exactly	PROVED; A.1–A.4
c -sequence formula (*), row comparison / flip form	PROVED; B1, B2
Sandwich property	PROVED; B3
Last step before δ_k has $c_t = k - 1$	PROVED; B4
Griggs–Ho Lemma 3.3(2), 3.4, 3.5, 3.6 (re-proved)	PROVED; B5–B8
$d_B(\lambda) \leq k^2 - k$ for all $\lambda \vdash T_k$	PROVED; B9
$D_B(T_k) = k^2 - k$ for $k \leq 8$; witness orbit for $k \leq 40$; lemma statements for $n \leq 30$	COMPUTER-VERIFIED, sanity only (not load-bearing); Section 5

3 Cited results versus our contribution

Cited (existing literature).

- J. R. Griggs and C.-C. Ho, *The cycling of partitions and compositions under repeated shifts*, Adv. Appl. Math. 21 (1998) 205–227 [1]. Theorem 3.7 there states $D_B(T_k) = k^2 - k$; its proof, via Proposition 3.2 and Lemmas 3.3–3.6, is the **source of the architecture of our upper bound** (Part B). Theorem 3.1 there gives the lower bound with the same witness $\lambda^{(k)}$ (its orbit is not written out there: “it is easy to see”). The convergence argument of 0.3 follows the idea of their Theorem 2.1. **Every lemma used from [1] is re-proved in full here**; in particular their Lemma 3.5 (“Continue this process ...”) is written as an explicit induction $I(m)$ in B7.
- The same value $D_B(T_k) = k^2 - k$ is proved by K. Igusa (1985) [2] and G. Etienne (1991) [3], and by H.-J. Bentz (1987) [4] (per [1]; these sources were not opened by the team).

Ours (the team’s contribution). The complete written proof below: the self-contained diagram description of B (0.1), the convergence and uniqueness of the cyclic partition for T_k (0.3, written out), the explicit orbit of the witness (Part A), and the full re-derivation of Griggs–Ho’s Proposition 3.2 and Lemmas 3.3–3.6 and the counting in their Theorem 3.7 (Part B). The mathematical result itself is the one in [1, 2, 3]; we do not claim it as ours.

4 Proof

Statement. For every $k \geq 1$, $D_B(T_k) = k^2 - k$ (no exceptions: $k = 1$ gives 0, $k = 2$ gives 2). LOWER bound (Part A): witness $\lambda^{(k)} = (k-1, k-1, k-2, \dots, 2, 1, 1)$ ($\lambda^{(1)} = (1)$), $d_B(\lambda^{(k)}) = k^2 - k$. UPPER bound (Part B): $d_B(\lambda) \leq k^2 - k$ for all $\lambda \vdash T_k$; the architecture follows Griggs–Ho (1998) Thm 3.7, with every lemma re-derived here in full (their Lemma 3.5 “continue this process” written as an explicit induction). Part 0.3 proves (self-contained) that δ_k is the only cyclic partition of T_k and every orbit reaches it, so no result of Cell 1 is assumed.

Notation. A partition is identified with its diagram $S = \{(i, j) : 1 \leq j \leq s, 1 \leq i \leq \lambda_j\}$ (column j = pile j , row i = level). Diagonal of a cell: $w(i, j) = i + j - 1$. Diagonal w has the w positions $(w, 1), (w-1, 2), \dots, (1, w)$; call the position $(w+1-j, j)$ “column j of diagonal w ”. $D_k :=$ diagram of δ_k = all cells with $w \leq k$ (column j of δ_k has height $k+1-j$, top cell on diagonal k).

Part 0. The shift on diagrams

0.1 Lemma. Let λ have s parts, diagram S . Define $S' = \text{rot}(S)$ by $(i, j) \mapsto (i-1, j+1)$ if $i \geq 2$, and $(1, j) \mapsto (j, 1)$.

- (a) rot preserves every diagonal and acts on diagonal w as the cyclic shift column $j \rightarrow j+1$ ($j < w$), column $w \rightarrow$ column 1.
- (b) S' has column 1 of height s and column $j+1$ of height $\lambda_j - 1$ ($1 \leq j \leq s$); these are exactly the parts of $B(\lambda)$ (plus zeros).
- (c) If $s \geq \lambda_1 - 1$ then S' is a Young diagram, so $\text{diagram}(B(\lambda)) = S'$.
- (d) If $s < \lambda_1 - 1$ then $\text{diagram}(B(\lambda))$ is obtained from S' by moving every cell (i, j) of S' with $i > s$ to $(i, j-1)$ (all such cells have $j \geq 2$); each moved cell drops from diagonal w to $w-1$, and at least one cell moves.

Proof. (a): $(i-1) + (j+1) - 1 = i + j - 1$; $(j, 1)$ and $(1, j)$ both lie on diagonal j . Column index $j \rightarrow j+1$, and $(1, w)$ (column w) $\rightarrow (w, 1)$ (column 1).

(b): the cells $(1, 1), \dots, (1, s)$ go to $(1, 1), \dots, (s, 1)$: column 1 of height s . The cells (i, j) , $2 \leq i \leq \lambda_j$, go to $(i-1, j+1)$: column $j+1$ of height $\lambda_j - 1$. $B(\lambda)$ has exactly the parts s and the positive $\lambda_j - 1$.

(c): column heights of S' are $s, \lambda_1 - 1 \geq \lambda_2 - 1 \geq \dots$; weakly decreasing iff $s \geq \lambda_1 - 1$; a weakly decreasing list of column heights is a Young diagram with the right parts.

(d): the diagram of the sorted partition has row i of length $r_i = \#\{\text{columns of } S' \text{ of height } \geq i\}$. For $i \leq s$, column 1 has height $\geq i$ and the columns $j \geq 2$ with height $\geq i$ form an initial segment $2..m$ (heights decreasing from column 2 on), so row i of S' is already $\{1..r_i\}$: unchanged. For $i > s$, column 1 has height $s < i$, so row i of S' is $\{2..r_i + 1\}$; in the sorted diagram it is $\{1..r_i\}$: every cell (i, j) with $i > s$ moves to $(i, j-1)$. Since $\lambda_1 - 1 > s$, the cell $(s+1, 2)$ exists: at least one move. $\square$

0.2 Lemma. $B(\delta_k) = \delta_k$, so δ_k is cyclic.

Proof. Parts $k-1, \dots, 1, 0 \rightarrow$ positive ones $k-1, \dots, 1$, plus new part $s = k$. $\square$

0.3 Lemma (convergence; Griggs–Ho Thm 2.1 argument, written out). For $n = T_k$ every orbit reaches δ_k , and δ_k is the only cyclic partition of T_k .

Proof. Energy $E(S) :=$ sum over cells of their diagonal index. Diagonal w has exactly w positions, so D_k is the set of all T_k positions on diagonals $1..k$. For $S \vdash T_k$ with $S \neq D_k$: $\#(\text{cells of } S \text{ on diagonals } > k) = \#(\text{holes of } S \text{ on diagonals } \leq k) =: h \geq 1$ and

$$E(S) - E(D_k) = \sum_{\text{cells on diag } w > k} (w - k) + \sum_{\text{holes on diag } w \leq k} (k - w) \geq h \geq 1.$$

By 0.1, one shift preserves E if no slide occurs and lowers E by ≥ 1 if a slide occurs (0.1(d)).

Claim: if $S \neq D_k$ then, for the index w defined next, a slide occurs at one of the steps $0, 1, \dots, w(w+1) - 1$ of the orbit of S . Let $W_0 =$ largest diagonal carrying a cell ($W_0 \geq k+1$) and $w_0 =$ smallest diagonal with a hole ($w_0 \leq k$). Let w be the largest index $< W_0$ whose diagonal has a hole (exists: $w_0 < W_0$). Diagonal $w+1$ carries a cell: if $w+1 = W_0$ by definition of W_0 ; otherwise diagonal $w+1$ has no hole (maximality of w), so it is full, hence non-empty. Fix a hole on diagonal w at column a and a cell on diagonal $w+1$ at column b . If no slide occurs during steps $0.. \tau - 1$, the diagram at time τ is $\text{rot}^\tau(S)$ (0.1(c)), in which (by 0.1(a)) that hole sits at column $a + \tau \pmod{w}$ and that cell at column $b + \tau \pmod{w+1}$. By CRT ($\gcd(w, w+1) = 1$) choose $\tau \in [0, w(w+1))$ with $a + \tau \equiv w \pmod{w}$ and $b + \tau \equiv 1 \pmod{w+1}$. At time τ , $(1, w)$ (column w of diagonal w) is empty, so the number of parts $s \leq w-1$ (row 1 is an initial segment); $(w+1, 1)$ (column 1 of diagonal $w+1$) is occupied, so $\mu_1 \geq w+1 > s+1$; by 0.1(d) the next step is a slide. So some slide occurs at a step $\leq \tau$.

Hence if the orbit never met δ_k , slides would occur infinitely often and E (a non-negative integer, never increasing) would decrease infinitely often: impossible. So $B^t(\lambda) = \delta_k$ for some t , and then for all larger t (0.2). If μ is cyclic, $B^p(\mu) = \mu$ ($p \geq 1$) gives $\mu = B^{mp}(\mu) = \delta_k$ for large m . $\square$

Part A. Lower bound: $d_B(\lambda^{(k)}) = k^2 - k$ for every $k \geq 1$

A.0 $k = 1$: $T_1 = 1$, the only partition is $(1) = \delta_1$, cyclic (0.2), $d_B = 0 = 1^2 - 1$. From now on $k \geq 2$.

A.1 Witness. $\lambda^{(k)} := (\lambda_1, \dots, \lambda_{k+1})$ with $\lambda_1 = k-1$, $\lambda_j = k+1-j$ ($2 \leq j \leq k$), $\lambda_{k+1} = 1$. It is weakly decreasing ($k-1 \geq k-1 \geq k-2 \geq \dots \geq 1 \geq 1$), positive, and sums to $(k-1) + (T_k - k) + 1 = T_k$. ($k = 2$: $(1, 1, 1)$; $k = 3$: $(2, 2, 1, 1)$.) Its diagram is D_k minus the cell $(k, 1)$ (= column 1 of diagonal k) plus the cell $(1, k+1)$ (= column $k+1$ of diagonal $k+1$): columns $2..k$ agree with δ_k , column 1 is one shorter, column $k+1$ is new of height 1.

A.2 Family. For $a \in \{1..k\}$, $b \in \{1..k+1\}$ let $S(a, b) := D_k \setminus \{\text{column } a \text{ of diagonal } k\} \cup \{\text{column } b \text{ of diagonal } k+1\}$ (a set of T_k cells). *Claim:* if $S(a, b)$ is the diagram of a partition μ , then (i) μ has $s = k - [a = k] + [b = k+1]$ parts and $\mu_1 = k - [a = 1] + [b = 1]$; (ii) if not ($a = k$ and $b = 1$) then $\text{diagram}(B(\mu)) = S(a+1 \bmod k, b+1 \bmod k+1)$ (residues taken in $\{1..k\}, \{1..k+1\}$); (iii) if $a = k$ and $b = 1$ then $B(\mu) = \delta_k$.

Proof. (i) $s =$ number of cells in row 1. Row-1 cells are $(1, j) =$ column j of diagonal j ; D_k has $(1, 1..k)$; the removed cell is in row 1 iff $a = k$ (it is $(k+1-a, a)$); the added cell $(k+2-b, b)$ is in row 1 iff $b = k+1$. Similarly $\mu_1 =$ number of cells in column 1: D_k has $(1..k, 1)$; removed cell in column 1 iff $a = 1$; added iff $b = 1$.

(ii) Since $k \geq 2$, $a = k$ excludes $a = 1$. $s \geq k-1$ always and $\mu_1 - 1 \leq k$; $s < \mu_1 - 1$ forces $s = k-1$ (so $a = k$, $b \neq k+1$) and $\mu_1 = k+1$ (so $b = 1$). So if not ($a = k$ and $b = 1$) then $s \geq \mu_1 - 1$, and by 0.1(c),(a) $\text{diagram}(B(\mu)) = \text{rot}(S(a, b))$; rot maps each full diagonal of D_k onto itself and shifts the hole and the extra cell by one column cyclically: $\text{rot}(S(a, b)) = S(a+1, b+1)$ (cyclically).

(iii) $a = k$, $b = 1$: $s = k-1$, $\mu_1 = k+1 > s+1$, so 0.1(d) applies. $\text{rot}(S(k, 1)) = S(1, 2)$ as a set: hole at $(k, 1)$, extra cell at $(k, 2)$. The cells of $\text{rot}(S(k, 1))$ in rows $> s = k-1$: rows $\geq k$. D_k has only $(k, 1)$ in rows $\geq k$, and it is the hole; the extra cell $(k, 2)$ is in row k . So exactly $(k, 2)$ moves to $(k, 1)$, filling the hole: result $D_k = \delta_k$. $\square$

A.3 Orbit. By A.1, $\lambda^{(k)} = S(1, k+1)$. Put $a_t = 1 + (t \bmod k)$, $b_t = 1 + ((t+k) \bmod (k+1))$. By A.2(ii) and induction, $\text{diagram}(B^t(\lambda^{(k)})) = S(a_t, b_t)$ for all $t \leq t^*$ where t^* is the least $t \geq 0$ with

$a_t = k$ and $b_t = 1$ (at every earlier step the hypothesis of (ii) holds; each $S(a_t, b_t)$ so produced is the diagram of the partition $B^t(\lambda^{(k)})$, so the hypothesis “ $S(a, b)$ is a diagram” holds by induction).

$a_t = k \iff t \equiv -1 \pmod{k}$; $b_t = 1 \iff t + k \equiv 0 \pmod{k+1} \iff t \equiv 1 \pmod{k+1}$. Since $\gcd(k, k+1) = 1$, the solutions form one class mod $k(k+1)$ (CRT). $t_0 = k^2 - k - 1$ is a solution: $k^2 - k - 1 \equiv -1 \pmod{k}$; mod $k+1$, $k \equiv -1$ so $k^2 - k - 1 \equiv 1 + 1 - 1 = 1$. For $k \geq 2$, $0 \leq t_0 < k(k+1)$, so $t^* = t_0 = k^2 - k - 1$.

By A.2(iii), $B^{k^2-k}(\lambda^{(k)}) = \delta_k$. For $0 \leq t \leq k^2 - k - 1$, $B^t(\lambda^{(k)})$ has diagram $S(a_t, b_t)$, which contains a cell on diagonal $k+1$, so it is not δ_k .

A.4 *Not cyclic before time $k^2 - k$.* Let $0 \leq t < k^2 - k$ and $\mu = B^t(\lambda^{(k)})$. If μ were cyclic, $B^p(\mu) = \mu$ for some $p \geq 1$, hence $\mu = B^{mp}(\mu)$ for all $m \geq 1$; choose $mp \geq k^2 - k - t$; then $B^{mp}(\mu) = B^{mp-(k^2-k-t)}(\delta_k) = \delta_k$ by A.3 and 0.2. So $\mu = \delta_k$, contradicting A.3. Hence $d_B(\lambda^{(k)}) = k^2 - k$ ($B^{k^2-k}(\lambda^{(k)}) = \delta_k$ is cyclic by 0.2). (This part uses no fact about which partitions of T_k are cyclic beyond 0.2.) Hence $D_B(T_k) \geq k^2 - k$ for all $k \geq 1$.

Part B. Upper bound $d_B(\lambda) \leq k^2 - k$ for $\lambda \vdash T_k$

(Architecture of Griggs–Ho Thm 3.7, all lemmas re-derived.) By 0.3, $d_B(\lambda) = \min\{t : B^t(\lambda) = \delta_k\}$ for $\lambda \vdash T_k$.

B1 (*c-sequence*). For $\lambda \vdash n$ with s parts let $c_u :=$ number of parts of $B^{u-1}(\lambda)$, $u \geq 1$ (so $c_1 = s$). By induction on t , the parts of $B^t(\lambda)$ are the positive numbers among $\{\lambda_j - t : j \leq s\} \cup \{c_u - (t - u) : 1 \leq u \leq t\}$. ($t = 0$ trivial; step: B subtracts 1 from each part, drops non-positive values, which is the same as subtracting 1 from every listed number and keeping positive ones, and adds the part $c_t = c_t - 0$.) Hence

$$c_{t+1} = \#\{j : \lambda_j \geq t + 1\} + \#\{u \leq t : c_u \geq t + 1 - u\}. \quad (*)$$

Write $f_{i,j} = [\lambda_j \geq i]$ and $e_{i,u} = [c_u \geq i - u]$ ($u < i$); then $c_i = \sum_j f_{i,j} + \sum_{u < i} e_{i,u}$ (“row i ”), and for fixed column (j or u) the entries are 1 on an initial segment of rows (monotone in i). Every $c_u \geq 1$ ($n \geq 1$), so $e_{i,i-1} = 1$.

B2 (*row comparison; Griggs–Ho “comparing rows”*). $c_{i+1} \leq c_i + 1$, with equality iff $f_{i+1,j} = f_{i,j}$ for all j and $e_{i+1,u} = e_{i,u}$ for all $u < i$.

Proof. Row $i+1$ consists of the entries $f_{i+1,j} \leq f_{i,j}$, $e_{i+1,u} \leq e_{i,u}$ ($u < i$) (monotonicity in the row index) and the new entry $e_{i+1,i} = 1$. So

$$c_{i+1} = 1 + (\text{sum of entries each } \leq \text{the corresponding entry of row } i) \leq 1 + c_i,$$

with equality iff all those entries are equal. □

(This is Griggs–Ho Prop. 3.2(1).) *Flip form:* call an old column (a λ -column j , or a c -column $u < i$) a *flip* at row i if its entry is 1 in row i and 0 in row $i+1$; for a c -column u this happens iff $u + c_u = i$ (“column u ends at row i ”), for a λ -column iff $\lambda_j = i$. With $L_i :=$ number of flips at row i , $c_{i+1} = c_i + 1 - L_i$. In particular $c_{i+1} = c_i + 1$ iff $L_i = 0$ (rows i and $i+1$ agree on all old columns), and $c_{i+1} = c_i$ iff $L_i = 1$ (they agree on all old columns except exactly one flip).

B3 (*Sandwich; Griggs–Ho Prop. 3.2(3)*). If $i < j$ and $c_i < x < c_j$ then there are $p < q$ with $i \leq p$, $q \leq j$, $q \geq p + 2$ and $(c_p, \dots, c_q) = (x - 1, x, \dots, x, x + 1)$.

Proof. Let p be the largest index in $[i, j]$ with $c_p \leq x - 1$ (exists: $p = i$ qualifies). Then $c_{p+1} \geq x$ (if $p + 1 < j$ by maximality; if $p + 1 = j$ since $c_j > x$), and $c_{p+1} \leq c_p + 1 \leq x$ by B2, so $c_p = x - 1$, $c_{p+1} = x$. All indices in (p, j) have $c \geq x$. Let q be the least index $> p + 1$ with $c_q \neq x$; $q \leq j$ since $c_j > x$. $c_q \geq x$ (as $q \leq j$ and either $q < j$ or $q = j$) and $c_q \leq c_{q-1} + 1 = x + 1$, so $c_q = x + 1$, $q \geq p + 2$. $\square$

B4 (*end of the orbit; Griggs-Ho Lemma 3.3(1)*). For $n = T_k$ and $t = d_B(\lambda) \geq 1$: $c_{t+1} = c_{t+2} = \dots = k$ and $c_t = k - 1$.

Proof. $B^t(\lambda) = \delta_k$ has k parts and is fixed (0.2), so $c_u = k$ for $u \geq t + 1$. Let $\mu = B^{t-1}(\lambda) \neq \delta_k$ (minimality of t , and δ_k cyclic). $B(\mu) = \delta_k$ contains the part c_t , so $c_t \leq k$; B2 gives $c_t \geq c_{t+1} - 1 = k - 1$. If $c_t = k$ then μ has k parts whose values minus 1 give $k - 1$ positive values equal to $k - 1, \dots, 1$ and one zero, i.e. $\mu = \delta_k$, contradiction. So $c_t = k - 1$. $\square$

B5 (*Griggs-Ho Lemma 3.3(2); proof written out*). For $n = T_k$ and $t = d_B(\lambda) \geq k + 1$, at least one holds: (i) $(c_p \dots c_q) = (k - 1, k, \dots, k, k + 1)$ for some $t - k \leq p < q \leq t - 1$, $q \geq p + 2$; (ii) $(c_p \dots c_q) = (k - 2, k - 1, \dots, k - 1, k)$ for some $t - k + 1 \leq p < q \leq t + 1$, $q \geq p + 2$.

Proof. By B4, $c_t = k - 1$ and $c_u = k$ for $u \geq t + 1$. Row t contains the k entries $e_{t,t-k}, \dots, e_{t,t-1}$ ($t - k \geq 1$), whose sum is $\leq c_t = k - 1$, so one of them is 0; $e_{t,t-1} = 1$ (B1), so there is $i \in [t - k, t - 2]$ with $e_{t,i} = 0$; take i maximal, so $e_{t,u} = 1$ for $u \in [i + 1, t - 1]$. Since $c_{t+1} = c_t + 1$, $L_t = 0$ (B2): $e_{t+1,i+1} = e_{t,i+1} = 1$. Hence $c_{i+1} \geq (t + 1) - (i + 1) = t - i$, and $e_{t,i} = 0$ gives $c_i \leq t - i - 1$; with $c_{i+1} \leq c_i + 1$ (B2): $c_i = t - i - 1$, $c_{i+1} = t - i$.

Case $i \geq t - k + 1$: $c_i = t - i - 1 \leq k - 2 < k - 1 < k = c_{t+1}$; B3 with $x = k - 1$ on $[i, t + 1]$ gives (ii) with $p \geq i \geq t - k + 1$, $q \leq t + 1$.

Case $i = t - k$: $c_{t-k} = k - 1$, $c_{t-k+1} = k$. If some $j \in [t - k + 2, t - 1]$ has $c_j \leq k - 2$: B3 with $x = k - 1$ on $[j, t + 1]$ gives (ii) with $p \geq j \geq t - k + 2$. If some such j has $c_j \geq k + 1$: B3 with $x = k$ on $[t - k, j]$ ($c_{t-k} = k - 1 < k < c_j$) gives (i) with $p \geq t - k$, $q \leq j \leq t - 1$. Otherwise $c_j \in \{k - 1, k\}$ for all $j \in [t - k + 2, t - 1]$, and $c_{t-k+1} = k$; so $c_u \leq k$ for all $u \in [t - k + 1, t - 1]$. Row t has exactly $c_t = k - 1$ ones and $e_{t,u} = 1$ for the $k - 1$ indices $u \in [t - k + 1, t - 1]$; so these are all the ones of row t . By B1 the parts of $\mu := B^{t-1}(\lambda)$ are the positive numbers among $\lambda_j - (t - 1)$ and $c_u - (t - 1 - u)$ ($u \leq t - 1$), and a column contributes a positive part iff its entry in row t is 1. So the parts of μ are $c_u - (t - 1 - u)$, $u \in [t - k + 1, t - 1]$, and their sum is $\leq \sum_{v=0}^{k-2} (k - v) = k + (k - 1) + \dots + 2 = T_k - 1 < T_k = |\mu|$, a contradiction. $\square$

B6 (*Griggs-Ho Lemma 3.4; proof written out*). For $\lambda \vdash n \geq 1$, if $(c_p, \dots, c_{p+m}) = (m - 2, m - 1, \dots, m - 1, m)$ then $p + m \leq n + 1$.

Proof. $c_p = m - 2 \geq 1$, so $m \geq 3$. Column p ends at row $p + m - 2$: $e_{p+m-2,p} = 1$, $e_{p+m-1,p} = 0$ (so also $e_{p+m,p} = 0$). For $u \in [p + 1, p + m - 1]$, $c_u = m - 1$, so column u has ones in rows $u + 1 \dots u + m - 1$, which contain row $p + m$: $e_{p+m,u} = 1$ ($m - 1$ ones). Row $p + m$ has $c_{p+m} = m$ ones, so exactly one further 1, in a column Z not in $[p, p + m - 1]$.

If Z is a λ -column j : $\lambda_j \geq p + m$, and $\lambda_j \leq n$, so $p + m \leq n$.

If Z is a c -column u_0 (then $u_0 \leq p - 1$): suppose $u_0 \geq 2$. Then $e_{p+m,u_0-1} = 0$ (the only ones of row $p + m$ are u_0 and $[p + 1, p + m - 1]$). $c_{p+m} = c_{p+m-1} + 1$, so $L_{p+m-1} = 0$ and row $p + m - 1$ agrees with row $p + m$ on old columns: $e_{p+m-1,u_0} = 1$, $e_{p+m-1,u_0-1} = 0$. $c_{p+m-1} = c_{p+m-2}$ ($= m - 1$, as $p + m - 2 \geq p + 1$), so $L_{p+m-2} = 1$, and column p is a flip at row $p + m - 2$; so it is the only one, and rows $p + m - 2$, $p + m - 1$ agree on the columns u_0 , $u_0 - 1$ (both $\neq p$): $e_{p+m-2,u_0-1} = 0$. So $c_{u_0} \geq (p + m) - u_0$ and $c_{u_0-1} \leq (p + m - 2) - (u_0 - 1) - 1 = p + m - u_0 - 2$, i.e.

$c_{u_0} \geq c_{u_0-1} + 2$, contradicting B2. Hence $u_0 = 1$, and $e_{p+m,1} = 1$ gives $c_1 \geq p + m - 1$; $c_1 = s \leq n$, so $p + m \leq n + 1$. $\square$

B7 (Griggs–Ho Lemma 3.5; proof written out, their “continue this process” made an induction). For $\lambda \vdash n \geq 1$: if $q \geq p+3$ and $(c_p..c_q) = (x-1, x, \dots, x, x+1)$ then either $p \leq x$ or there is a pattern $(c_{p'}..c_{q'}) = (x'-1, x', \dots, x', x'+1)$ with $x' \leq x$, $2 \leq q' - p' < q - p$ and $p - x \leq p' < q' \leq p + 1$.

Proof. Assume $p \geq x + 1$. The x entries $e_{p,p-x}, \dots, e_{p,p-1}$ ($p - x \geq 1$) lie in row p , which has $c_p = x - 1$ ones, so one is 0; $e_{p,p-1} = 1$, so take p' maximal in $[p - x, p - 2]$ with $e_{p,p'} = 0$; then $e_{p,u} = 1$ for $u \in [p' + 1, p - 1]$. $c_{p+1} = c_p + 1$ gives $L_p = 0$: $e_{p+1,u} = 1$ for $u \in [p' + 1, p]$ ($u = p$ is the new entry). As in B5, $e_{p+1,p'+1} = 1$ and $e_{p,p'} = 0$ give, with B2, $c_{p'} = y - 1$ and $c_{p'+1} = y$ where $y := p - p' \in [2, x]$.

Invariant $I(m)$ ($m \geq 1$): $p + m \leq q - 1$; $c_{p'+1} = \dots = c_{p'+m} = y$; and $e_{p+m,u} = 1$ for all $u \in [p' + m, p + m - 1]$. $I(1)$ holds (its first clause because $q \geq p + 3$). Note $c_{p'+m} = y$ means column $p' + m$ ends at row $p' + m + y = p + m$, so it is a flip at row $p + m$.

Step. Assume $I(m)$. (a) If $p + m = q - 1$: $c_q = c_{q-1} + 1$ forces $L_{q-1} = 0$, but column $p' + m$ is a flip at row $q - 1$: contradiction. So $p + m \leq q - 2$ (which is the first clause of $I(m + 1)$), hence $c_{p+m} = c_{p+m+1} = x$ and $L_{p+m} = 1$; the unique flip is column $p' + m$, so row $p + m + 1$ agrees with row $p + m$ on the other old columns: $e_{p+m+1,u} = 1$ for $u \in [p' + m + 1, p + m]$ ($u = p + m$ is the new entry). (b) $e_{p+m+1,p'+m+1} = 1$ gives $c_{p'+m+1} \geq y$, and B2 gives $c_{p'+m+1} \leq c_{p'+m} + 1 = y + 1$. If $c_{p'+m+1} = y + 1$, stop with $q' := p' + m + 1$; else $I(m + 1)$ holds.

Termination: by (a), $I(m)$ forces $m \leq q - 2 - p$, so the process stops at some $m \leq q - p - 2$, giving the pattern $(c_{p'}, \dots, c_{q'}) = (y - 1, y, \dots, y, y + 1)$ with $q' - p' = m + 1 \in [2, q - p - 1]$, i.e. $2 \leq q' - p' < q - p$. Put $x' = y \leq x$; $p' \geq p - x$. Finally $q' \leq p + 1$: the indices $p' + 1..q' - 1$ carry the value y and q' carries $y + 1$; if $q' \geq p + 2$ then indices p and $p + 1$ would both lie in $[p' + 1, q' - 1]$ ($p' \leq p - 2$), forcing $c_p = c_{p+1} = y$, but $c_p = x - 1 \neq x = c_{p+1}$. $\square$

B8 (Griggs–Ho Lemma 3.6; proof written out). If $(c_p, c_{p+1}, c_{p+2}) = (x - 1, x, x + 1)$ then $p \leq x$.

Proof. Suppose $p > x$, i.e. $p - 1 > x - 1 = c_p$. Row p contains the $p - 1$ entries $e_{p,1}, \dots, e_{p,p-1}$ (all counted in c_p), so at most $c_p < p - 1$ of them are 1: some $e_{p,i} = 0$ with $i \leq p - 1$; since $e_{p,p-1} = 1$ (B1), $i \leq p - 2$. Take i maximal: $e_{p,i} = 0$, $e_{p,i+1} = 1$. By B2 equality ($c_{p+1} = c_p + 1$, then $c_{p+2} = c_{p+1} + 1$): $e_{p+1,i+1} = e_{p,i+1} = 1$ and $e_{p+2,i+1} = e_{p+1,i+1} = 1$. So $c_{i+1} \geq (p + 2) - (i + 1) = p - i + 1$, while $e_{p,i} = 0$ gives $c_i \leq p - i - 1$. Then $c_{i+1} \geq c_i + 2$, contradicting B2. $\square$

B9 (assembly; Griggs–Ho proof of Thm 3.7, counting written out). Let $\lambda \vdash T_k$, $t = d_B(\lambda)$.

- $k = 1, 2$: exhaustive by hand: $k = 1$ only (1), $d = 0$. $k = 2$: (2, 1) fixed; (3) $\rightarrow$ (2, 1); (1, 1, 1) $\rightarrow$ (3) $\rightarrow$ (2, 1); $\max 2 = k^2 - k$.
- $k \geq 3$ and $t \leq k$: $t \leq k < k^2 - k$.
- $k \geq 3$, $t \geq k + 1$: by B5 (i) or (ii) holds.
 - (ii) with $q = p + k$: then $p = t - k + 1$ (forced by the ranges $p \geq t - k + 1$, $q \leq t + 1$), and B6 ($m = k$) gives $p + k \leq n + 1$, so $t = p + k - 1 \leq n = k(k + 1)/2 \leq k^2 - k$ (since $k \geq 3$).

- otherwise $q - p \leq k - 1$ and $p \geq t - k$. Set $P_0 = (p, q, x)$ ($x = k$ or $k - 1$; $x \leq k$). While the current pattern (p_i, q_i, x_i) has $q_i - p_i \geq 3$ and $p_i > x_i$, apply B7 to get $(p_{i+1}, q_{i+1}, x_{i+1})$ with $x_{i+1} \leq x_i \leq k$, $q_{i+1} - p_{i+1} \leq q_i - p_i - 1$, $p_{i+1} \geq p_i - x_i \geq p_i - k$. The length $q_i - p_i$ starts $\leq k - 1$, drops by ≥ 1 per step, and stays ≥ 2 , so at most $k - 3$ steps occur. The process stops at some $m \leq k - 3$ with $p_m \leq x_m \leq k$ (either B7's first alternative, or $q_m - p_m = 2$ and B8). Hence $p_0 \leq p_m + mk \leq k + (k - 3)k = k^2 - 2k$, and $t \leq p_0 + k \leq k^2 - k$.

So $d_B(\lambda) \leq k^2 - k$ for all $\lambda \vdash T_k$, i.e. $D_B(T_k) \leq k^2 - k$. With Part A: $D_B(T_k) = k^2 - k$ for every $k \geq 1$. $\square$

Conclusion. Established here (written proofs, no citation used as proof): $D_B(T_k) = k^2 - k$ for every $k \geq 1$; lower bound by the explicit witness $\lambda^{(k)}$ (Part A), upper bound by Part B (Griggs–Ho architecture, all lemmas re-derived). The most delicate step is B7 (index bookkeeping). The same result is proved in Griggs–Ho Thm 3.7, Igusa (1985), Etienne (1991).

5 How to verify (computations; sanity only, not load-bearing)

No step of the proof above uses a computation; the scripts below only check instances and prove nothing for all k . They are stdlib-only Python 3 and are shipped in the folder `code/` of this submission. Run from the submission folder (measured with Python 3.14.0 on macOS, Darwin 23.4.0; wall-clock times from `/usr/bin/time -p`):

- `python3 code/witness.py 40` — simulates B on $\lambda^{(k)}$, checks $d_B(\lambda^{(k)}) = k^2 - k$ and that $B^t(\lambda^{(k)})$ has diagram $S(a_t, b_t)$ of A.3 for $t < k^2 - k$, for $k = 1..40$. Expected output: `k=1..40 checked, failures: 0`. Runtime 1.91 s.
- `python3 code/dbt.py 8` — exhaustive $D_B(T_k)$ for $k = 1..8$: values 0, 2, 6, 12, 20, 30, 42, 56 = $k^2 - k$ (number of maximisers 1, 1, 1, 3, 16, 65, 293, 1267); e.g. $k = 3$: only (2, 2, 1, 1). Expected: each line shows `D_B=` equal to `k^2-k=`. Runtime 0.64 s.
- `python3 code/gh_lemmas.py 30 80` — checks the statements of B5 ($n = T_k \leq 28$, all 4411 partitions with $d_B \geq k + 1$), B6, B7, B8 (all partitions of $n \leq 30$, $c_1..c_{80}$). Expected output: `partitions checked 28628 n<= 30 seq length 80 violations L3.6: 0 L3.5: 0 and L3.4 violations: 0 L3.3(2): cases 4411 violations 0`. Runtime 3.84 s.

All arithmetic is exact integer arithmetic; total runtime well under the 10-minute limit.

6 Limitations

- The result is complete for the cell as stated: both bounds are proved for every $k \geq 1$, with no exceptions. Nothing is claimed beyond $n = T_k$ (e.g. no statement about which partitions attain the maximum for general k , and nothing about non-triangular n).
- The upper-bound architecture is that of Griggs–Ho [1]; the proof here re-derives every lemma, and its correctness rests on the written argument, which the reader should check (the delicate step is B7). The computer checks of Section 5 are finite-range sanity checks only.
- The references [2, 3, 4] were not opened by the team; their content is reported as described in [1] and the survey [5].

- Editorial changes relative to the accepted proof text (no change to the mathematics): a reference to an internal assumption of the team was removed from the first line of Part B (0.3 already proves what is needed); in B7 the bound $p + m \leq q - 1$, which step (a) of the proof derives for $I(m + 1)$, is stated as part of the invariant $I(m)$; code paths point to the shipped `code/` folder.

References

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